

Kevin_Hooman_Assignment3.1

May 25, 2026

```
[27]: # Homework 3 _ Feature Group
```

```
[ ]: #%pip install --force-reinstall 'sagemaker>=2.200,<3'
import boto3
import sagemaker
from sagemaker.session import Session
from sagemaker import get_execution_role
import pandas as pd
import numpy as np
import time
from time import gmtime, strftime, sleep
from sagemaker.feature_store.feature_group import FeatureGroup
import pprint
```

```
sagemaker.config INFO - Not applying SDK defaults from location:
/etc/xdg/sagemaker/config.yaml
```

```
sagemaker.config INFO - Not applying SDK defaults from location:
/home/sagemaker-user/.config/sagemaker/config.yaml
```

```
[3]: print("sagemaker version:", sagemaker.__version__) # expect something like 2.
      ↪254.x
```

```
sagemaker version: 2.257.3
```

```
[4]: original_boto3_version = boto3.__version__
      ↪%pip install -q 'boto3>1.17.21'
```

```
[5]: region = boto3.Session().region_name
      boto_session = boto3.Session(region_name=region)

      sagemaker_client = boto_session.client("sagemaker", region_name=region)
      featurestore_runtime = boto_session.client("sagemaker-featurestore-runtime",
      ↪region_name=region)

      feature_store_session = Session(
          boto_session=boto_session,
          sagemaker_client=sagemaker_client,
          sagemaker_featurestore_runtime_client=featurestore_runtime,
```

```
)

default_s3_bucket_name = feature_store_session.default_bucket()
prefix = "sagemaker-featurestore-homework-3-1"
print("Region :", region)
print("Bucket :", default_s3_bucket_name)
```

```
Region : us-east-1
Bucket : sagemaker-us-east-1-697214847177
```

```
[6]: role = get_execution_role()
print("Role :", role)
```

```
Role : arn:aws:iam::697214847177:role/LabRole
```

```
[7]: HOUSING_PATH = "housing.csv"
GMAPS_PATH = "housing_gmaps_data_raw.csv"

housing = pd.read_csv(HOUSING_PATH)
gmaps = pd.read_csv(GMAPS_PATH)

print("housing:", housing.shape, " gmaps:", gmaps.shape)
housing.head()
```

```
housing: (20640, 10) gmaps: (12590, 30)
```

```
[7]:
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	\
0	-122.23	37.88	41.0	880.0	129.0	
1	-122.22	37.86	21.0	7099.0	1106.0	
2	-122.24	37.85	52.0	1467.0	190.0	
3	-122.25	37.85	52.0	1274.0	235.0	
4	-122.25	37.85	52.0	1627.0	280.0	

	population	households	median_income	median_house_value	ocean_proximity
0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	496.0	177.0	7.2574	352100.0	NEAR BAY
3	558.0	219.0	5.6431	341300.0	NEAR BAY
4	565.0	259.0	3.8462	342200.0	NEAR BAY

```
[8]: gmaps_slim = (gmaps[["latitude", "longitude", "neighborhood-political",
↪ "postal_code"]])
        .drop_duplicates(subset=["latitude", "longitude"]))

df = housing.merge(gmaps_slim, on=["latitude", "longitude"], how="left")

# neighborhood label
df = df[df["neighborhood-political"].notna()].copy()
```

```

print("Rows after join + neighborhood filter:", len(df))
print("Distinct neighborhoods          :", df['neighborhood-political'].
      ↪nunique())

```

Rows after join + neighborhood filter: 9000
 Distinct neighborhoods : 1306

```

[9]: # one-hot encode ocean_proximity with all 5 categories enforced
ALL_OCEAN = ["<1H OCEAN", "INLAND", "ISLAND", "NEAR BAY", "NEAR OCEAN"]
df["ocean_proximity"] = pd.Categorical(df["ocean_proximity"],
      ↪categories=ALL_OCEAN)

ocean_oh = pd.get_dummies(df["ocean_proximity"]).astype(int)
ocean_oh.columns = (ocean_oh.columns.str.lower()
      ↪.str.replace("<1h ocean", "lt_1h_ocean",
      ↪regex=False)
      ↪.str.replace(" ", "_"))
df = pd.concat([df, ocean_oh], axis=1)

OH_COLS = list(ocean_oh.columns)
print("One-hot columns:", OH_COLS)

```

One-hot columns: ['lt_1h_ocean', 'inland', 'island', 'near_bay', 'near_ocean']

```

[10]: # bedrooms_per_household at the row level
df["bedrooms_per_household"] = df["total_bedrooms"] / df["households"]

# Impute: where the ratio is NaN (because total_bedrooms was null),
# fill with the mean ratio for that row's postal code
df["bedrooms_per_household"] = (df.
      ↪groupby("postal_code")["bedrooms_per_household"]
      ↪.transform(lambda s: s.fillna(s.mean()))

# Safety net: a postal code with zero non-null ratios would leave residual NaNs
df["bedrooms_per_household"] = df["bedrooms_per_household"].fillna(
      ↪df["bedrooms_per_household"].mean()
      ↪)

print("Residual nulls in bedrooms_per_household:", df["bedrooms_per_household"].
      ↪isna().sum())

```

Residual nulls in bedrooms_per_household: 0

```

[11]: agg = (df.groupby("neighborhood-political")
      ↪.agg(**{c: (c, "max") for c in OH_COLS},
      ↪median_house_value = ("median_house_value", "mean"),
      ↪median_house_age_raw = ("housing_median_age", "mean"),
      ↪total_households = ("households", "mean"),

```

```

        bedrooms_per_household = ("bedrooms_per_household", "mean"))
    .reset_index()
    .rename(columns={"neighborhood-political": "neighborhood"}))

# Cap median_house_value at 500,000
agg["median_house_value"] = agg["median_house_value"].clip(upper=500000)

# Round households up to integer
agg["total_households"] = np.ceil(agg["total_households"]).astype(int)

# Bucket the averaged median house age into 10year ranges
def age_bucket(avg_age):
    lo = int(avg_age // 10) * 10
    return f"{lo}-{lo+9}"

agg["median_house_age"] = agg["median_house_age_raw"].apply(age_bucket)
agg = agg.drop(columns=["median_house_age_raw"])

# Final column order
neighborhood_features = agg[
    ["neighborhood"] + OH_COLS +
    ["median_house_value", "median_house_age", "total_households",
    ↪ "bedrooms_per_household"]
].copy()

print("Feature group shape:", neighborhood_features.shape)
print("Any nulls?", neighborhood_features.isna().any().any())
neighborhood_features.head()

```

Feature group shape: (1306, 10)

Any nulls? False

```

[11]:
      neighborhood  lt_1h_ocean  inland  island  near_bay  \
0              28 Palms         1       0       0         0
1      Acorn Industrial         0       0       0         1
2      Adams Hill           1       0       0         0
3  Agua Mansa Industrial Corridor  0       1       0         0
4              Al Tahoe         0       1       0         0

      near_ocean  median_house_value  median_house_age  total_households  \
0              0      222200.000000          20-29              923
1              0       81300.000000          50-59              147
2              0     250733.333333          30-39              494
3              0     112300.000000          10-19              516
4              0     109180.000000          20-29              249

      bedrooms_per_household

```

```

0          1.017335
1          1.659864
2          1.034649
3          1.102713
4          1.641739

```

```
[12]: #Building the Feature Group
```

```
[13]: def cast_object_to_string(frame):
    for col in frame.columns:
        if frame.dtypes[col] == "object":
            frame[col] = frame[col].astype("str").astype("string")

cast_object_to_string(neighborhood_features)

record_identifier_feature_name = "neighborhood"
event_time_feature_name       = "event_time"

current_time_sec = float(round(time.time()))
neighborhood_features[event_time_feature_name] = pd.Series(
    [current_time_sec] * len(neighborhood_features), dtype="float64"
)

neighborhood_features.dtypes
```

```
[13]: neighborhood          str
lt_1h_ocean                int64
inland                     int64
island                     int64
near_bay                   int64
near_ocean                 int64
median_house_value         float64
median_house_age           str
total_households           int64
bedrooms_per_household     float64
event_time                 float64
dtype: object
```

```
[16]: for col in ["neighborhood", "median_house_age"]:
    neighborhood_features[col] = neighborhood_features[col].astype(object)

print(neighborhood_features.dtypes) # verify
```

```

neighborhood          object
lt_1h_ocean          int64
inland                int64
island                int64
near_bay              int64

```

```

near_ocean                int64
median_house_value        float64
median_house_age          object
total_households          int64
bedrooms_per_household    float64
event_time                float64
dtype: object

```

```

[17]: neighborhood_feature_group_name = "neighborhood-feature-group-" + \
    ↳ strftime("%d-%H-%M-%S", gmtime())
    print("FeatureGroup name:", neighborhood_feature_group_name)

    neighborhood_feature_group = FeatureGroup(
        name=neighborhood_feature_group_name,
        sagemaker_session=feature_store_session,
    )

    neighborhood_feature_group.
    ↳ load_feature_definitions(data_frame=neighborhood_features)

```

FeatureGroup name: neighborhood-feature-group-25-14-30-23

```

[17]: [FeatureDefinition(feature_name='neighborhood',
    feature_type=<FeatureTypeEnum.STRING: 'String'>, collection_type=None),
    FeatureDefinition(feature_name='lt_1h_ocean',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='inland',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='island',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='near_bay',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='near_ocean',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='median_house_value',
    feature_type=<FeatureTypeEnum.FRACTIONAL: 'Fractional'>, collection_type=None),
    FeatureDefinition(feature_name='median_house_age',
    feature_type=<FeatureTypeEnum.STRING: 'String'>, collection_type=None),
    FeatureDefinition(feature_name='total_households',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='bedrooms_per_household',
    feature_type=<FeatureTypeEnum.FRACTIONAL: 'Fractional'>, collection_type=None),
    FeatureDefinition(feature_name='event_time',
    feature_type=<FeatureTypeEnum.FRACTIONAL: 'Fractional'>, collection_type=None)]

```

```

[18]: def wait_for_feature_group_creation_complete(fg):
    status = fg.describe().get("FeatureGroupStatus")
    while status == "Creating":

```

```

        print("Waiting for FeatureGroup creation...")
        time.sleep(5)
        status = fg.describe().get("FeatureGroupStatus")
        if status != "Created":
            raise RuntimeError(f"Failed to create FeatureGroup {fg.name}")
        print(f"FeatureGroup {fg.name} successfully created.")

neighborhood_feature_group.create(
    s3_uri=f"s3://{default_s3_bucket_name}/{prefix}",
    record_identifier_name=record_identifier_feature_name,
    event_time_feature_name=event_time_feature_name,
    role_arn=role,
    enable_online_store=True,
)

wait_for_feature_group_creation_complete(neighborhood_feature_group)

```

Waiting for FeatureGroup creation...

Waiting for FeatureGroup creation...

Waiting for FeatureGroup creation...

Waiting for FeatureGroup creation...

FeatureGroup neighborhood-feature-group-25-14-30-23 successfully created.

```
[19]: neighborhood_feature_group.describe()
```

```

[19]: {'FeatureGroupArn': 'arn:aws:sagemaker:us-east-1:697214847177:feature-
group/neighborhood-feature-group-25-14-30-23',
'FeatureGroupName': 'neighborhood-feature-group-25-14-30-23',
'RecordIdentifierFeatureName': 'neighborhood',
'EventTimeFeatureName': 'event_time',
'FeatureDefinitions': [{'FeatureName': 'neighborhood',
'FeatureType': 'String'},
{'FeatureName': 'lt_1h_ocean', 'FeatureType': 'Integral'},
{'FeatureName': 'inland', 'FeatureType': 'Integral'},
{'FeatureName': 'island', 'FeatureType': 'Integral'},
{'FeatureName': 'near_bay', 'FeatureType': 'Integral'},
{'FeatureName': 'near_ocean', 'FeatureType': 'Integral'},
{'FeatureName': 'median_house_value', 'FeatureType': 'Fractional'},
{'FeatureName': 'median_house_age', 'FeatureType': 'String'},
{'FeatureName': 'total_households', 'FeatureType': 'Integral'},
{'FeatureName': 'bedrooms_per_household', 'FeatureType': 'Fractional'},
{'FeatureName': 'event_time', 'FeatureType': 'Fractional'}],
'CreationTime': datetime.datetime(2026, 5, 25, 14, 30, 27, 789000,
tzinfo=tzlocal()),
'OnlineStoreConfig': {'EnableOnlineStore': True},
'OfflineStoreConfig': {'S3StorageConfig': {'S3Uri': 's3://sagemaker-us-

```

```

east-1-697214847177/sagemaker-featurestore-homework-3-1',
  'ResolvedOutputS3Uri': 's3://sagemaker-us-east-1-697214847177/sagemaker-
featurestore-homework-3-1/697214847177/sagemaker/us-east-1/offline-
store/neighborhood-feature-group-25-14-30-23-1779719427/data'},
  'DisableGlueTableCreation': False,
  'DataCatalogConfig': {'TableName':
'neighborhood_feature_group_25_14_30_23_1779719427',
  'Catalog': 'AwsDataCatalog',
  'Database': 'sagemaker_featurestore'}}},
  'ThroughputConfig': {'ThroughputMode': 'OnDemand'},
  'RoleArn': 'arn:aws:iam::697214847177:role/LabRole',
  'FeatureGroupStatus': 'Created',
  'OnlineStoreTotalSizeBytes': 0,
  'ResponseMetadata': {'RequestId': 'f902bd14-5442-4579-bf02-7f604324dcc8',
  'HTTPStatusCode': 200,
  'HTTPHeaders': {'x-amzn-requestid': 'f902bd14-5442-4579-bf02-7f604324dcc8',
  'strict-transport-security': 'max-age=47304000; includeSubDomains',
  'x-frame-options': 'DENY',
  'content-security-policy': "frame-ancestors 'none'",
  'cache-control': 'no-cache, no-store, must-revalidate',
  'x-content-type-options': 'nosniff',
  'content-type': 'application/x-amz-json-1.1',
  'content-length': '2136',
  'date': 'Mon, 25 May 2026 14:31:48 GMT'},
  'RetryAttempts': 0}}

```

```

[20]: neighborhood_feature_group.ingest(
    data_frame=neighborhood_features,
    max_workers=3,
    wait=True,
)

```

```

[20]: IngestionManagerPandas(feature_group_name='neighborhood-feature-
group-25-14-30-23', feature_definitions={'neighborhood': {'FeatureName':
'neighborhood', 'FeatureType': 'String'}, 'lt_1h_ocean': {'FeatureName':
'lt_1h_ocean', 'FeatureType': 'Integral'}, 'inland': {'FeatureName': 'inland',
'FeatureType': 'Integral'}, 'island': {'FeatureName': 'island', 'FeatureType':
'Integral'}, 'near_bay': {'FeatureName': 'near_bay', 'FeatureType': 'Integral'},
'near_ocean': {'FeatureName': 'near_ocean', 'FeatureType': 'Integral'},
'median_house_value': {'FeatureName': 'median_house_value', 'FeatureType':
'Fractional'}, 'median_house_age': {'FeatureName': 'median_house_age',
'FeatureType': 'String'}, 'total_households': {'FeatureName':
'total_households', 'FeatureType': 'Integral'}, 'bedrooms_per_household':
{'FeatureName': 'bedrooms_per_household', 'FeatureType': 'Fractional'},
'event_time': {'FeatureName': 'event_time', 'FeatureType': 'Fractional'}},
sagemaker_fs_runtime_client_config=<botocore.config.Config object at
0x7f7b7bdf73e0>, sagemaker_session=<sagemaker.session.Session object at

```



```
0x7f7b7cea0110>, max_workers=3, max_processes=1, profile_name=None,
_async_result=<multiprocessing.pool.MapResult object at 0x7f7b79656420>,
_processing_pool=<pool ProcessPool(ncpus=1)>, _failed_indices=[])
```

```
[21]: def fetch(neighborhood_name):
      resp = featurestore_runtime.get_record(
          FeatureGroupName=neighborhood_feature_group_name,
          RecordIdentifierValueAsString=neighborhood_name,
      )
      print(f"=== {neighborhood_name} ===")
      pprint.pp(resp["Record"])
      print()
      return resp

_ = fetch("Brooktree")
```

```
=== Brooktree ===
[{'FeatureName': 'neighborhood', 'ValueAsString': 'Brooktree'},
 {'FeatureName': 'lt_1h_ocean', 'ValueAsString': '1'},
 {'FeatureName': 'inland', 'ValueAsString': '0'},
 {'FeatureName': 'island', 'ValueAsString': '0'},
 {'FeatureName': 'near_bay', 'ValueAsString': '0'},
 {'FeatureName': 'near_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'median_house_value', 'ValueAsString': '257400.0'},
 {'FeatureName': 'median_house_age', 'ValueAsString': '0-9'},
 {'FeatureName': 'total_households', 'ValueAsString': '1438'},
 {'FeatureName': 'bedrooms_per_household',
  'ValueAsString': '1.0747777761942723'},
 {'FeatureName': 'event_time', 'ValueAsString': '1779718688.0'}]
```

```
[22]: _ = fetch("Fisherman's Wharf")
```

```
=== Fisherman's Wharf ===
[{'FeatureName': 'neighborhood', 'ValueAsString': "Fisherman's Wharf"},
 {'FeatureName': 'lt_1h_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'inland', 'ValueAsString': '0'},
 {'FeatureName': 'island', 'ValueAsString': '0'},
 {'FeatureName': 'near_bay', 'ValueAsString': '1'},
 {'FeatureName': 'near_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'median_house_value', 'ValueAsString': '500000.0'},
 {'FeatureName': 'median_house_age', 'ValueAsString': '50-59'},
 {'FeatureName': 'total_households', 'ValueAsString': '250'},
 {'FeatureName': 'bedrooms_per_household', 'ValueAsString': '1.268'},
 {'FeatureName': 'event_time', 'ValueAsString': '1779718688.0'}]
```

```
[23]: _ = fetch("Los Osos")
```

```

=== Los Osos ===
[{'FeatureName': 'neighborhood', 'ValueAsString': 'Los Osos'},
 {'FeatureName': 'lt_1h_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'inland', 'ValueAsString': '0'},
 {'FeatureName': 'island', 'ValueAsString': '0'},
 {'FeatureName': 'near_bay', 'ValueAsString': '0'},
 {'FeatureName': 'near_ocean', 'ValueAsString': '1'},
 {'FeatureName': 'median_house_value', 'ValueAsString': '221612.5'},
 {'FeatureName': 'median_house_age', 'ValueAsString': '10-19'},
 {'FeatureName': 'total_households', 'ValueAsString': '612'},
 {'FeatureName': 'bedrooms_per_household',
  'ValueAsString': '1.0478845404823531'},
 {'FeatureName': 'event_time', 'ValueAsString': '1779718688.0'}]

```

```

[24]: batch = featurestore_runtime.batch_get_record(Identifiers=[{
        "FeatureGroupName": neighborhood_feature_group_name,
        "RecordIdentifiersValueAsString": ["Brooktree", "Fisherman's Wharf", "Los_
↪Osos"],
    }])

rows = []
for rec in batch["Records"]:
    rows.append({f["FeatureName"]: f["ValueAsString"] for f in rec["Record"]})

pd.DataFrame(rows).set_index("neighborhood")

```

```

[24]:          lt_1h_ocean inland island near_bay near_ocean \
neighborhood
Brooktree          1      0      0      0      0
Fisherman's Wharf  0      0      0      1      0
Los Osos           0      0      0      0      1

          median_house_value median_house_age total_households \
neighborhood
Brooktree          257400.0      0-9      1438
Fisherman's Wharf  500000.0      50-59      250
Los Osos           221612.5      10-19      612

          bedrooms_per_household      event_time
neighborhood
Brooktree          1.0747777761942723  1779718688.0
Fisherman's Wharf          1.268  1779718688.0
Los Osos           1.0478845404823531  1779718688.0

```

```
[ ]:
```